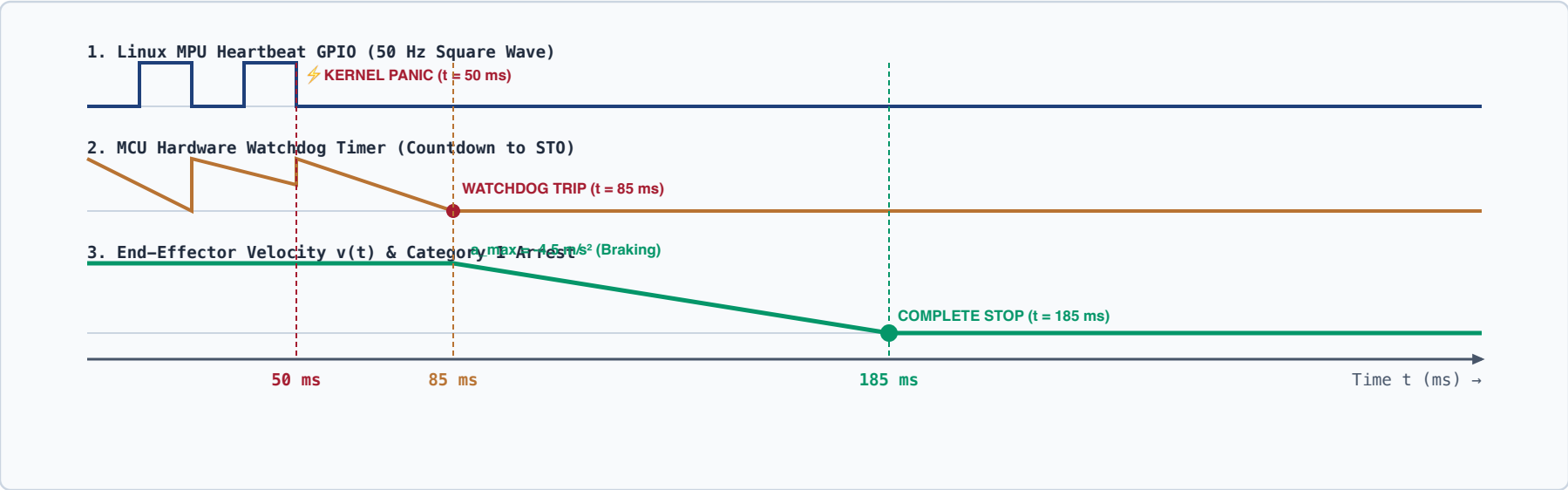


CROSS-LAYER SEEDED FAULT INJECTION & DYNAMIC ARREST WAVEFORM

Live telemetry: Linux MPU kernel crash at $t = 50\text{ ms}$ → Watchdog trip at $t = 85\text{ ms}$ → Safe arrest at $t = 185\text{ ms}$ ($d_{\text{stop}} = 5.2\text{ cm} < 12.0\text{ cm}$)



✓ PHYSICAL VERIFICATION: BOUNDED STOPPING MARGIN PROOF

Verified Stopping Margin:
 $d_{\text{stop}} = d_{\text{reaction}} + d_{\text{braking}} = 1.6\text{ cm} + 3.6\text{ cm}$

Conclusion: Even under total MPU hardware failure, the Reflex Brain brings the robot to a complete rest within physical clearance envelopes.